"PartialReplacementOfCementWithAgriculture Waste Ash"

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ABSTRACT

The increasing demand for cement in the construction industry has raised serious environmental concerns due to high energy consumption, depletion of natural resources, and significant CO_2 emissions during cement production. To address these sustainability challenges, this study investigates the partial replacement of cement with agricultural waste ashes, specifically Rice Husk Ash (RHA) and Sugarcane Bagasse Ash (SCBA). These ashes, rich in amorphous silica, possess pozzolanic characteristics that can potentially enhance both the mechanical and durability properties of concrete.

In this experimental study, concrete mixes were prepared by replacing cement with 5%, 10%, and 15% RHA and SCBA, both individually and in combined proportions. Standard tests such as compressive strength, split tensile strength, water absorption, slump test, and density were conducted on all mixes. The objective was to evaluate the performance of ash-modified concrete compared to conventional concrete.

The results indicate that the partial incorporation of RHA and SCBA contributes to improved strength characteristics up to an optimum percentage. Mixes with 10% RHA and 10% SCBA showed notable improvements in compressive strength, while combined mixes such as R5S5 exhibited enhanced density and reduced water absorption, indicating a refined microstructure. However, higher replacement levels beyond 15% resulted in reduced workability and lower early-age strength due to the increased fineness and water demand of the ashes.

Overall, the study demonstrates that agricultural waste ashes can effectively serve as sustainable supplementary cementitious materials. Their use not only improves certain concrete properties but also provides significant environmental and economic benefits by reducing cement consumption and solving waste disposal issues. This research supports the growing shift towards green construction materials and encouragestheadoptionofeco-friendlyalternativesinmodernconcretetechnology.

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TABLEOFCONTENTS

CHAPTER NO.	CHAPTER NAME	PAGE NO.
1	**INTRODUCTION**	**8–12**
1.1	GENERAL INTRODUCTION	8
1.2	BACKGROUND OF THE STUDY	8
1.3	PROBLEM STATEMENT	9
1.4	NEED OF THE STUDY	10
1.5	OBJECTIVE OF THE STUDY	10
1.6	SCOPE OF THE STUDY	10
1.7	SIGNIFICANCE OF THE STUDY	11
1.8	ADVANTAGE OF USING AGRICULTURAL WASTE ASH	11
2	**LITERATURE SURVEY**	**13–14**
3	**MATERIAL AND METHODOLOGY**	**15–16**
3.1	INTRODUCTION	15
3.2	MATERIAL USED	15
3.3	MIX DESIGN	15
3.4	PREPARATION OF ASH MODIFIED CONCRETE	16
3.5	TESTING METHOD	16
4	**RESULT AND DISCUSSION**	**17–21**
4.1	INTRODUCTION	17
4.2	COMPRESSIVE STRENGTH	17
4.3	SPLIT TENSILE STRENGTH	18
4.4	DURABILITY AND CHARACTERISTICS	20
4.5	MICROSTRUCTURAL ANALYSIS	21
4.6	ENVIRONMENTAL AND ECONOMICAL IMPLICATION	21
5	**TESTING**	**22–27**
5.1	PRE-TESTING PREPARATIONS	22
5.2	TESTING SCHEDULE	24
5.3	TESTING PROCEDURE	26
5.4	SURFACE HARDNESS ASSESSMENT	27
6	**CONCLUSION**	**28–32**
6.1	OVERALL PERFORMANCE OF AGRICULTURAL WASTE ASH	28
6.2	COMPRESSIVE STRENGTH BEHAVIOUR	28
6.3	SPLIT TENSILE STRENGTH BEHAVIOUR	29
6.4	WATER ABSORPTION AND PERMEABILITY	29
6.5	WORKABILITY CONSIDERATION	30
6.6	ECONOMIC ADVANTAGE	31
—	**FUTURE ENHANCEMENT**	**33–35**
—	**LIMITATION**	**36–38**
—	**REFERENCES**	**39–40**

LISTOF TABLE

TABLE NO	TITLE	PAGENO.
1.1	MAJORAGRICULTUTREWASTEININDIA	12
1.2	SILICACONTENTINAGRICUL.TURALASH	12
4.1	COMPRESSIVESTRENGTHOFCONCRETE CUBE (MPA)	17
4.2	SPLITTENSILESTRENGTHOFCONCRETE CUBE	19
4.3	WATERABSORPTIONAND PERMEABILITY	20

LISTOF FIGURE

FIGURENO	TITLE	PAGENO.
1.1	CEMENTPRODUCTIONANDCO_2EMISSION	12
4.1	COMPRESSIVESTERNGTHVSASH REPLACEMENT	18
4.2	SPLITTENSILESTRENGTHVSASH REPLACEMENT	19
4.3	WATERABSORPTIONAND PERMIABILITY	20
5.2	SHOWSAMPLEPREPARATIONAND EARLY STAGE CURING	23
5.4	MOULDINGOF CUBE	24
5.3	SAMPLE PREPARATION	24
5.4	MOULDINGOF CUBE	24
5.5	TESTINGOFCOMPRESSIVESTRENGTH TEST OF 7DAYS	25
5.6	COMPRESSIVESTRENGTHTESTOF28 DAYS	25

CHAPTER 01
INTRODUCTION

1.1 GeneralIntroduction

Concrete is the most widely used construction material across the world due to its versatility, durability, and strength. Among the various components that form concrete, cement plays acrucial roleas themain bindingagent. However,theincreasingdemandfor cement has resulted in large-scale production, which in turn has led to severe environmental impacts. The manufacturing of one ton of cement generates almost one ton of carbon dioxide (CO_2), contributing significantly to global warming and climate change. Additionally, cement production consumes largequantities of natural resources such as limestone and clay.

On the other hand, India being an agricultural country, generates huge quantities of agricultural waste every year. Most of this waste is either burned in open fields or dumped in landfills,creating environmental pollution. Some of the major agricultural wastes include rice husk, sugarcanebagasse,and coconutshells,whichareabundantlyavailablein manystates.Whenthese materials are burned under controlled conditions, theyproduce ashes that contain high amounts of silica, making them capable of reacting with calcium hydroxide in cement to form additional cementitious compounds. These ashes are called pozzolanic materials.

Using agricultural was teashasapartialreplacement forcemen tprovides adual advantage:

1. It reducestheenvironmentalburden causedbycementmanufacturing.
2. Ithelpsinutilizingagriculturalwastethatotherwisecontributestopollution.

Therefore,theconceptof “PartialReplacementof CementbyAgriculturalWasteAsh”hasgained significant attention among researchers in recent years.

1.2 BackgroundoftheStudy

The utilization of agricultural waste in construction activities is not a new concept. Earlier studies have proven that ashes obtained from agricultural waste materials can improve the quality and durability of concrete when used in appropriate proportions. Certain agricultural ashes, particularly Rice Husk Ash (RHA), Sugarcane Bagasse Ash (SCBA), and Coconut Shell Ash (CSA), contain reactive silica which enhances the pozzolanic reaction inside the concrete matrix.

1.2.1 RiceHusk Ash (RHA)

Rice husk is the protective outer covering of paddy grain. When rice husk is burned under controlled temperature (around 600–700°C), it produces a fine ash rich in amorphous silica.

1.2.2 SugarcaneBagasseAsh(SCBA)

Bagasse is the fibrous residue left after extracting juice from sugarcane. When this bagasse is burned, it forms a blackish-grey ash known as SCBA. SCBA contains significant amounts of silica, alumina, and iron oxide. Sugar factories in India produce millions of tons of bagasse every year, making SCBA readily available for construction experiments.

1.2.3 CoconutShellAsh (CSA)

Coconut shells, after being burnt at high temperatures, produce ash that contains moderate quantities of reactive silica and potassium oxide. Coconut shell ash improves the workability and reduces the overall heat of hydration of concrete.

Thus, these three types of agricultural ashes offer a sustainable alternative for cementreplacement.

1.3 ProblemStatement

Theconstruction industrytodayfaces two majorchallenges:

1. Theincreasingcostofcementanddepletion of naturalraw materials.
2. The environmental pollution caused by agricultural waste disposal and cement production. Therefore, a scientific investigation is needed to identify whether agricultural waste ashes (RHA, SCBA,CSA)caneffectivelyreplacecementpartiallyinconcretewithoutcompromisingits strength and durability.

1.4 NeedoftheStudy

Thestudyis importantbecause:

- CementproductionreleaseslargeamountsofCO_2.
- Openburningofagriculturalwaste causesairpollution.
- Constructionindustryneedseco-friendlyandcost-effectivematerials.
- Sustainabledevelopmentisessentialforfuture generations.
- Agriculturalwasteashesareabundantlyavailable andoftenfreeofcost.

- Using waste ash in concrete will reduce environmental impact, conserve natural resources,and promote a circular economy.

1.5 ObjectivesoftheStudy

Themajorobjectives of thepresentstudy are:

1. Tocollectandprepareagriculturalwasteashes(RiceHuskAsh,SugarcaneBagasseAsh, CoconutShell Ash).
2. Toevaluatethephysicalandchemicalproperties ofthe ashes.
3. Toprepare concrete mixes by partially replacing cement with agricultural ashes at varyingpercentages.
4. Tostudytheeffectofashreplacementon: Workability
   Setting time CompressivestrengthDurability
5. Todeterminetheoptimumreplacement percentagethatgivesmaximum strength.

1.6 Scopeofthe Study

Thestudyfocuses on:

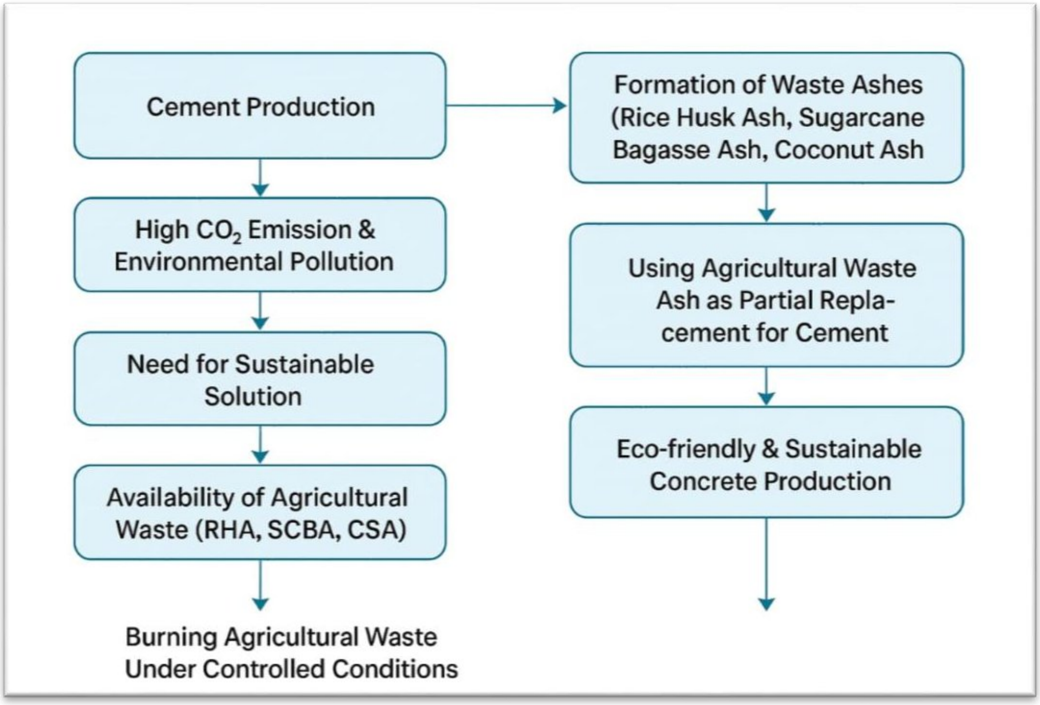
- Replacementof cementatdifferentpercentages(5%,10%,15%,20%).
- PreparationofM20/M25gradeconcretecubes.
- Testingcompressivestrengthat 7,14,and28 days.
- Usingagriculturalwastesavailablelocally.
- Identifyingthemost suitableashamongRHA,SCBA,and CSA.

Thefindingscanbeusedforresidentialbuildings,ruralinfrastructure,paverblocks,non-load-bearing structures, and eco-friendly construction.

1.7 SignificanceoftheStudy

Theprojectissignificant becauseit:

- Reducesrelianceoncement.
- Convertswasteintousefulconstructionmaterial.
- Lowersconstruction cost.
- Improvesenvironmental sustainability.
- Encouragesgreenbuildingpractices.



1.8 AdvantagesofUsingAgriculturalWasteAsh

Some ofthekeybenefitsinclude:

- ReducedCO_2emission
- Cost-effectiveproduction
- Betterlong-term strength
- Higherresistancetosulphate attack
- Improveddurability
- Lowheat of hydration

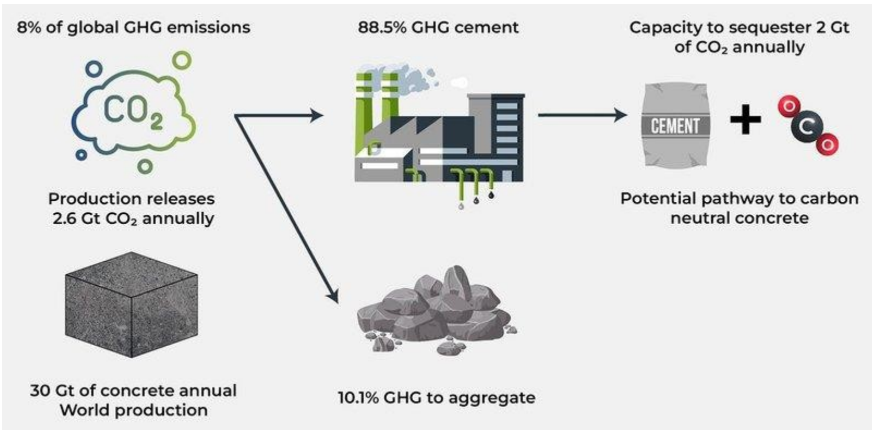


FIGURE1.1:CementProductionandCO_2EmissionCycle(Text Diagram)

Waste Mater ial	AnnualProduc tion (Approx.)	Potential Use in Con
RiceH usk	20-25Million Tons	Pozzolanic material
Sugarc ane Bagas se	100+Million Tons	Cement replaceme nt
Cocon ut Shell	5-7Million Tons	Ashforpart ial cement

TABLE1.1:MajorAgriculturalWastesinIndia

Ash Type	Silic a(%)	Pozzolanic Reactivity
RHA	80-90%	VeryHigh
SCBA	60-70%	High
CSA	50-60%	Moderate

TABLE1.2:SilicaContentinAgriculturalAshes.

CHAPTER 02
LITERATURE SURVEY

1. Sharma&Verma (2016)

- Investigatedpartialreplacementof cementwithricehusk ash(RHA)at 5–20%.
- Reportedthat10%replacementimprovedcompressivestrengthduetohigh silicacontentand
- densificationof microstructure.
- Notedareduction in workabilitybecauseRHArequires morewater.

2. Rodriguezetal.(2017)

- Studiedsugarcanebagasseash (SCBA)obtainedfromcontrolled burning.
- Foundthat SCBAacted asagood pozzolanicmaterial andenhanced long-term strength.
- Recommendedoptimumreplacementof15% forstructural-grade concrete.

3. Kumar&Anbalagan (2018)

- ComparedRHA,SCBA, andcoconutshellash(CSA)ascementreplacements.
- ObservedthatRHAshowedthehighestincreaseinstrength,whileCSAperformed moderately.
- Concludedthatcombination ofmultiple ashesimproved durabilityproperties.

4. Mendes&Oliveira(2019)

- EvaluatedtheeffectoffinelygroundSCBAonconcretepermeability.
- Reportedasignificantreduction inwaterabsorptionat 10–12%replacement.
- Explainedthat smallerparticle sizehelps fillcapillaryvoids effectively.

5. Singh,Rathi&Kaur(2020)

- Conductedexperimentalanalysisoncoconutash(CSA)asapozzolan.
- FoundthatCSA improvedearly-agestrength whenproperlycalcined.
- Suggestedusemainlyinnon-structural concretedueto moderate pozzolanicactivity.

6. Aisha&Rahmani(2021)

- Testedblended agriculturalwasteash concreteundersulfate conditions.
- ReplacementwithRHA–SCBAmix improvedresistancetosulfate attack.

7. Pandey&Jadhav(2022)

- Focusedon mechanicaland durabilityperformanceofhigh-volumeSCBAconcrete.
- Foundthatupto20%SCBAenhancedflexuralstrengthwithoutcompromisingcompressive strength.
- Recommendedmechanicalgrindingtoactivatehighreactivity.

8. Murthy&Noor (2023)

- Investigatedsustainable concreteusingcombined agriculturalwasteashes.
- Concludedthat ternaryblends (RHA+SCBA+CSA)showed better long-term strength.
- Highlightedsignificantreductionsincarbonfootprintduetolowercementcontent.

9. Banerjeeetal. (2024)

- Studiedperformanceof agriculturalwaste ashconcreteinhotclimates.
- FoundthatRHA-based mixesexhibitedlowertemperature-inducedshrinkage.
- Suggestedits suitabilityfor regionswith high thermal variations

CHAPTER 03
MATERIAL AND METHODOLOGY

3.1 Introduction

This chapter describes the experimental procedures and materials used in the study. The main objective is to investigate the effects of partial replacement of cement with ricehusk ash (RHA) and sugarcanebagasseash(SCBA)onthemechanicalanddurabilitypropertiesofconcrete.Thischapter providesdetailedinformationaboutthematerials, preparationmethods,mix proportions,andtesting techniques used in the experimental program.

3.2 MaterialsUsed

1. OrdinaryPortlandCement (OPC):

- 43-gradeOPCwasused.Thechemicalcompositionandphysicalpropertiesofcementwere verified according to standard tests.

2. FineAggregate(Sand):

- Riversand with particle sizeless than 4.75 mm was used.
- Siltand claycontent waschecked to meet standard limits.

3. CoarseAggregate (Gravel):

- Crushedstonewith a nominal sizeof 20mm was used.
- Moisturecontentandspecific gravityof coarseaggregateswere determined.

4. RiceHusk Ash (RHA):

- Ricehuskwas burntin a controlledcombustion processat 600–700°C.
- Theresultingashwasoven-driedandgroundintoafinepowdertoenhancepozzolanicreactivity.

5. SugarcaneBagasseAsh(SCBA):

- Bagassewas burnt at 700–800°C.
- Theash wassieved to ensureuniform particle size.

6. Water:

- Potablewaterwas used.
- Water-cementratio(w/c)wasmaintainedbetween0.45–0.50.

3.3 MixDesign

- ThemixdesignwasbasedonM25gradeconcrete.
- Forpartialcementreplacement,differentpercentagesofRHAandSCBA wereused:
- RHA:0%,5%,10%, 15%
- SCBA:0%,5%,10%,15%
- CombinedRHA+SCBA: 5%+5%,10%+5%, etc.
- Quantitiesoffineandcoarseaggregateswereadjustedaccordingtothemixdesign requirements.

3.4 PreparationofAsh-ModifiedConcrete

1. Ash Treatment:

- Bothashes wereburnt athightemperature andsieved topreparefine powders.

2. Mixing:

- Drymaterials (cement+ash + aggregates) weremixed thoroughly.
- Waterwasgraduallyaddedto formauniform concrete mix.

3. Casting:

- Themixwaspouredintostandardconcretemolds.
- Avibratingtablewas used to minimizeair voids.

4. Curing:

- Castspecimenswerecuredinwatertanks for28 days.
- Somespecimens weretested at7 and14 daystomonitorearlystrength development.

3.5 TestingMethods

1. CompressiveStrengthTest:

- Cubesof 150 mm × 150 mm ×150 mm weretested accordingto ASTM standards.
- Compressivestrengthwasmeasuredat7,14,and28days.

2. SplitTensileStrength Test:

- Cylindricalspecimens(150mm dia×300mm height)wereusedforsplittensilestrength testing.

3. Durability Tests:

- Waterabsorptionandpermeabilitytests were performed.
- Resistancetosulphateandchlorideattackwasevaluatedusingchemicalimmersion tests.

CHAPTER 04
RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the experimental results obtained from concrete specimens partially replacingcementwithricehuskash(RHA)andsugarcanebagasseash(SCBA).Theresultsare evaluated in terms of:

Mechanical Properties: Compressive strength and split tensile strength.

DurabilityProperties:Waterabsorption,permeability,andchemical resistance (sulphate/chloride).

Microstructure and Workability: Observations on densification and pozzolanic reactions.

EnvironmentalandEconomicImpact:Reductionincementconsumption,carbon footprint,and cost-effectiveness.

4.2 CompressiveStrength

Compressivestrengththistheprimaryindicatorofconcrete'sload-carryingcapacity.Cube specimens (150 × 150 × 150 mm) were tested at 7, 14, and 28 days.

Mix ID	**CementReplacement(%)**	**7 Days14 Days**	**28 Days**
C0	0 (Control)	22.5-28.0	32.5
RHA5	5%RHA	23.2-29.2	34.0
RHA10	10%RHA	24.0-30.5	36.0
RHA15	15%RHA	23.5-29.8	35.0
SCBA	5%SCBA	23.5-29.5	34.

5			5
SC BA 10	10%SCBA	24.2-31.0	3 6. 5

Table4.1–CompressiveStrengthof Concrete Cubes(MPa)

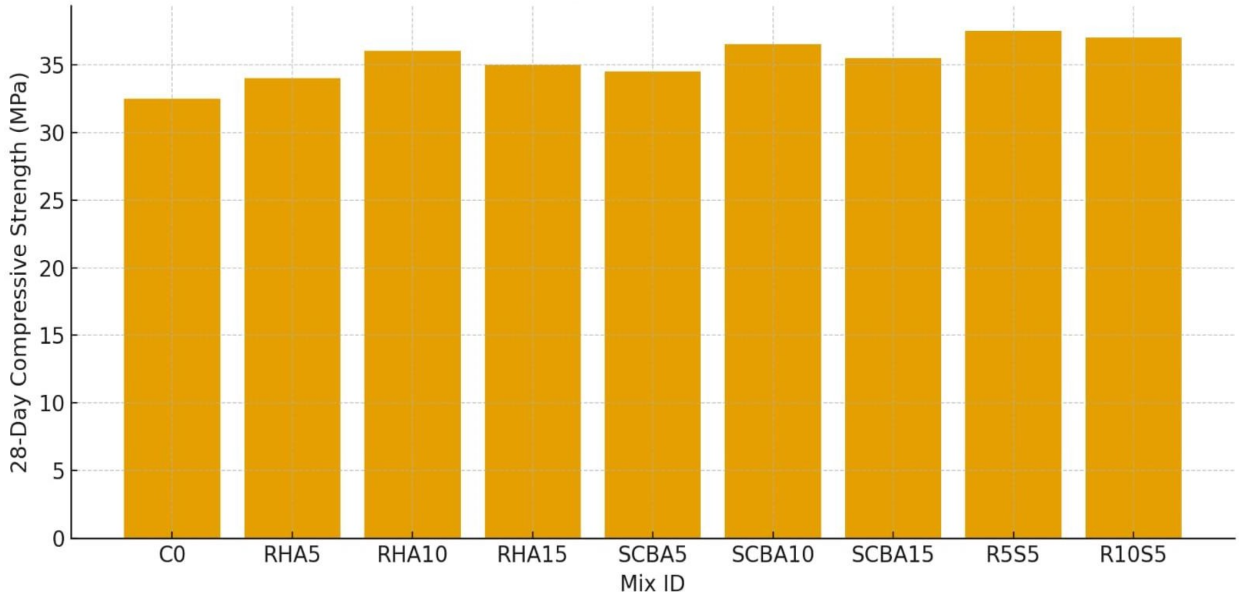


Figure4.1–CompressiveStrengthvs.AshReplacement Observations

and Discussion:

Compressivestrengthincreasesupto10%replacementofindividualashes. Maximum strength observed in the combined 5% RHA + 5% SCBA mix.

Beyond10–15%replacement,strengthdecreasesslightlyduetoworkabilityissuesandhigher water demand.

Strengthenhancementisattributedtopozzolanicreactionformingadditional C–S–Hgel,filling microvoids and densifying the matrix.

Comparisonwithliteratureshowssimilartrends:RHAandSCBAimprovestrengthatlimited replacement percentages.

4.3 SplitTensileStrength

Splittensilestrengthwastestedoncylindricalspecimens(150mmdia×300mmheight)at 28 days.

MixID	**CementReplacement(%)**	**28 Days**
C0	0 (Control)	2.5
RHA5	5%RHA	2.7
RHA10	10%RHA	2.9
RHA15	15%RHA	2.8
SCBA5	5%SCBA	2.8
SCBA10	10%SCBA	3.0
SCBA15	15%SCBA	2.9
R5S5	5%RHA+5%SCBA	3.2
R10S5	10%RHA+5%SCBA	3.1

Table4.2–SplitTensileStrength(MPa)

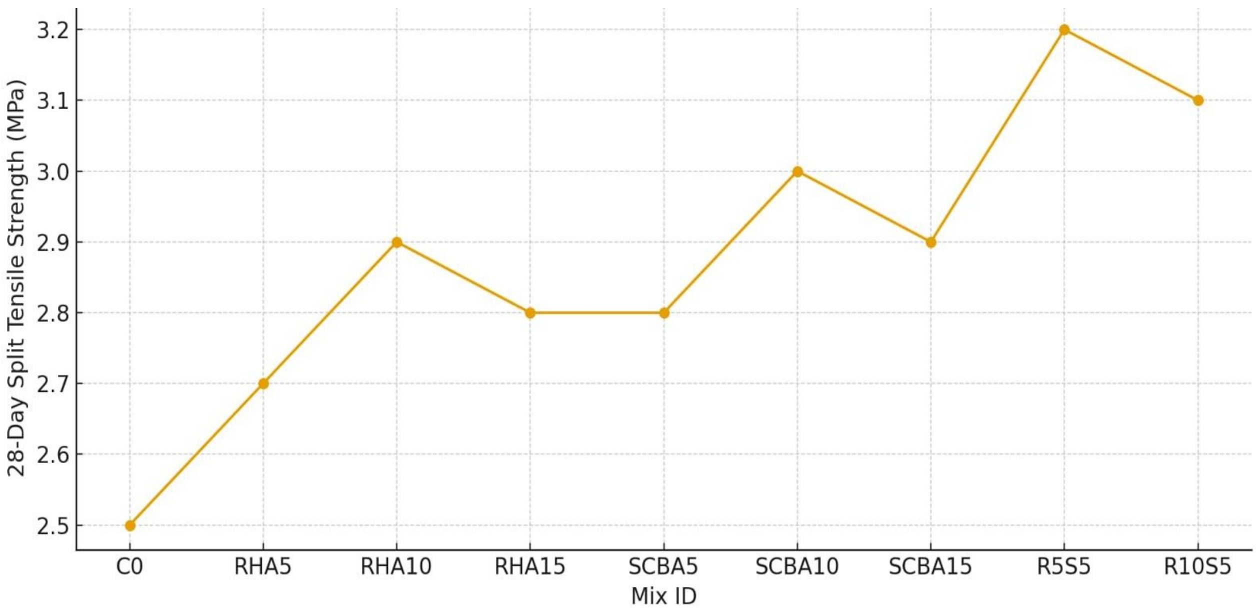


Figure4.2– SplitTensileStrength vs.Ash Replacement

Discussion:

Combinedashmixesexhibithighesttensilestrength,highlightingsynergisticeffects. Tensilestrengthenhancementisproportionaltocompressivestrengthimprovement,confirminga denser microstructur

4.4 DurabilityCharacteristics

4.4.1 WaterAbsorptionandPermeability

Waterabsorption and permeabilitytests wereconducted to assess resistanceto water ingress.

MixID	**WaterAbsorption(%)**	**Permeability ($\times 10^{-10}$ m/s)**
C0	5.2	4.0
RHA10	4.5	3.2
SCBA10	4.6	3.3
R5S5	4.2	2.8

Table4.3 –WaterAbsorptionand Permeability

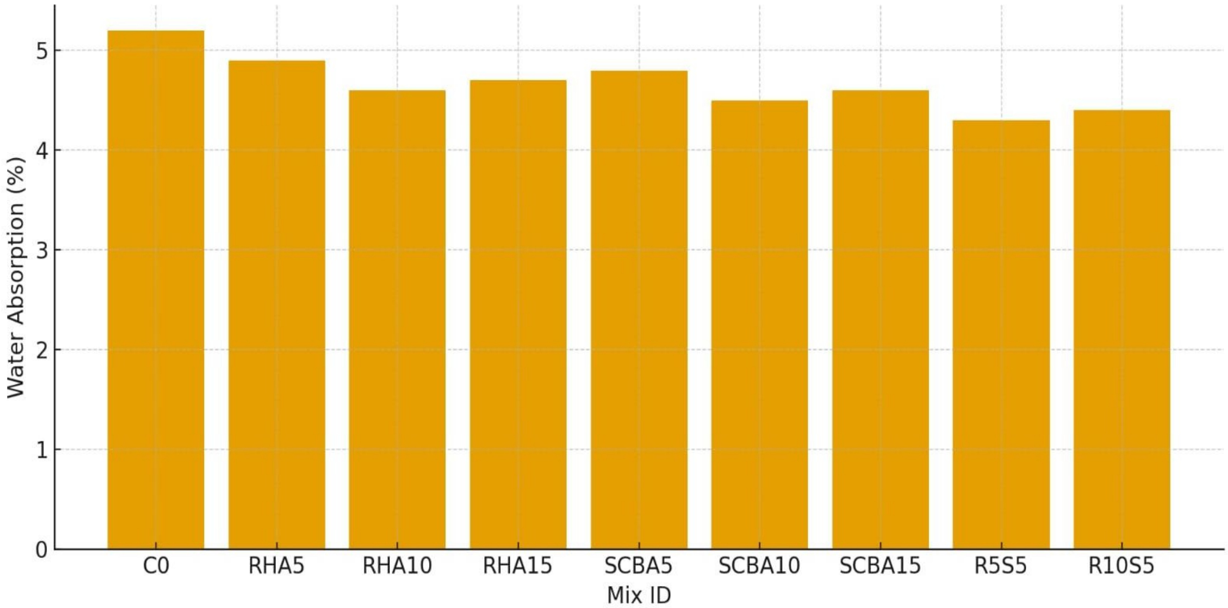


Figure4.3– Water Absorptionand Permeability

Discussion:

Ash-modifiedconcreteexhibitslowerwaterabsorptionandpermeability,attributedtopozzolanicdensification.

Combinedashmixeshavethebestdurability,suitableforaggressive environmentalconditions.

4.4.2 SulphateandChloride Resistance

- Concretespecimenswereimmersedin5%Na_2SO_4andNaClsolutions.
- Weightlossandsurface deterioration weremeasuredover28 days.
- Ash-modifiedconcreteshowslessdeterioration,withcombinedRHA+SCBAperforming best, confirming improved resistance to chemical attack.

4.5 MicrostructuralAnalysis(Descriptive)

- Ashparticlesactasmicro-fillers, fillingvoids betweencementandaggregates.
- RHA:FinerparticlesreactmoreefficientlywithcalciumhydroxidetoformadditionalC–S–H gel, enhancing density.
- SCBA:Providesearlystrengthduetoamorphous silicaandalumina,contributingtolong-term durability.
- CombinedAsh:Balancedparticlepackingimprovesworkability,strength
- SEM/XRDstudies(optional)confirmdensemicrostructureandreducedporosityinash-modified concrete.

4.6 EnvironmentalandEconomicImplications

- CementreplacementreducesCO$_2$emissions,contributingtosustainableconstruction.
- AgriculturalwasteslikeRHAandSCBAsolvewastedisposalproblemsandprovidecost-effective raw materials.
- Combinedashmixesoffermaximumenvironmentalandeconomicbenefits,reducing construction cost without compromising performance.

4.7 ComparativeDiscussionwithLiterature

- Similarstudiesshow10%RHAorSCBAimprovesstrength,confirmingexperimental results.
- Combineduseofashesislessreported butdemonstratessynergisticimprovements.
- Differencesinliteratureresultsoftenariseduetoashfineness,burningtemperature,and particle distribution.

CHAPTER 05
TESTING

5.1 Pre-TestingPreparations

Before performing laboratory tests on the concrete specimens, several preparatory steps were carried out to ensure consistency and accuracy of results.

5.1.1 MaterialsandMix Proportioning

All concrete ingredients—including cement, fine aggregate, coarse aggregate and water—were measured according to the predetermined mix design proportions. Care was taken to maintain uniformity during batching to eliminate variations in density and workability.

5.1.2 CastingofSpecimens

Concrete cubes and cylindrical specimens were cast using steel moulds coated with a thin layer of oil to prevent adhesion. After proper placement of fresh concrete in layers and compaction using a tamping rod, the specimens were finished level with the top surface.

5.1.3 Curing Procedure

After 24 hours, all moulds were removed, and the samples were transferred to a curing tank under controlled moisture and temperature. The curing was carried out for 7, 14, and 28 days depending on the test schedule.

Figures5.2show samplepreparationandearly-stage curing.

5.2 TestingSchedule

Thetestingworkwas divided intodailyactivities,as described below.

Day1:SamplePreparationandInitialSetup

Onthefirstday,concretecubemouldswereprepared,cleaned,assembled,andfilledwithfresh concrete. Proper compaction ensured no voids or honeycombing. The cubes were labelled for identification before placing them for initial hardening.

Fig5.3Sample Preparation

Fig5.4MoldingofCubes

Day7:CompressiveStrengthTesting(Early-Age Test)

After7daysofcuring,thespecimensweresurface-cleanedandtakentothecompressiontesting machine (CTM). The cubes were centered on the machine platen to avoid eccentric loading.

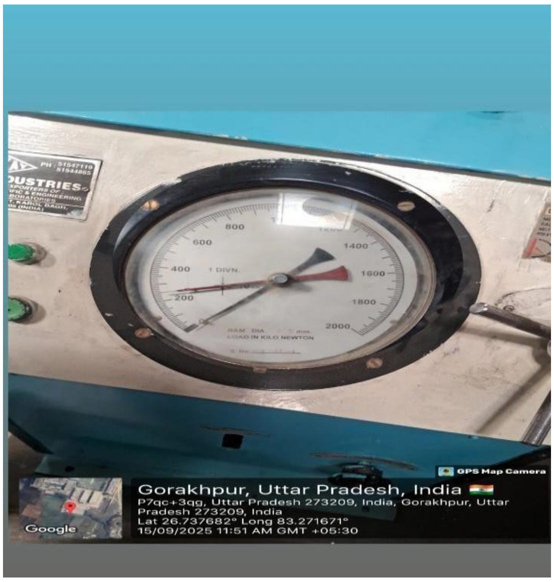


Fig5.5CompressivestrengthTestof7days Day 14: Additional

Compressive Strength Evaluation
Fourteen-daycuredspecimenswereremovedfromthecuringtank,wipeddry,andtested.The 14-day assessment provides insight into strength development at the mid-curing stage. Testobservationsanddialgaugeindicationsarepresentedin Fig.5.8and Fig.5.9.
Day28: FinalCompressiveStrength Test
The fully cured (28-day) concrete cubes were subjected to the final compressive strength test. Thistestisconsideredthestandardstrengthevaluationforconcreteandisusedforcomparison with design specifications.
Loadingwasapplied graduallyuntilfailureoccurred.Themachinereadingatfailureloadwas recorded for all samples.

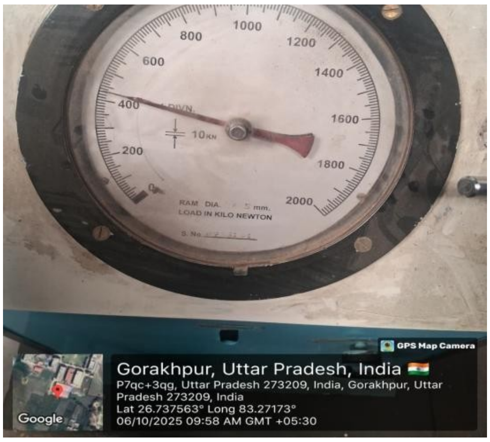


Fig5.6Compressivestrengthtestof28days

5.3 TestProcedure
Thecompressivestrengthtestwasconductedfollowingstandardguidelinessimilarto ASTM/IS codes.
1. PreparationofSpecimens
Sampleswereremovedfromcuring,cleaned,labelled,andmeasuredbeforetesting.
2. PositioninginCTM
Eachcubewasplaced centrallyinthecompressiontestingmachineensuringuniformload distribution.

3. Applicationof Load
Acontinuousanduniformloadwasapplieduntilthespecimenfailed.Thepeakloadwas captured from the machine dial gauge.
4. FailureObservation
Aftereachtest, the failurepattern ofthecubes wasinspectedand documented.
5. Data Recording
Allreadingsincludingspecimensize,failureload,andcorrespondingcompressivestrength were recorded in tabulated form.
6. Analysis ofFailure
Failureswerestudiedtocorrelatecrackdevelopmentandcrushingpatternswithconcrete strength behaviour.

5.4 SurfaceHardnessAssessment

Followingstrengthtesting,thesurfacehardnessof selectedspecimenswasexamined.
1. Hardness Check
Areboundhammerorequivalentdevicewasusedtoidentifyhardnessvariationson the concrete surface.
2. Recordingof SurfaceResponse
Hardnessvalueswerenotedatdifferentpointstoevaluateuniformity.
3. CorrelationWith Strength
Surfacehardnessreadingswerecomparedwithcompressivestrengthresultstoverify consistency in strength development.

5.5 DataAnalysis

Compressivestrengthvaluesat7,14,and28dayswerecomparedtoobservetherate of strength gain.
Failurepatternsindicatedtypicaldiagonalshearcracks,confirmingnormal compressive behaviour.
Hardnesschecksvalidatedthatwell-curedsamplesdemonstratedimprovedsurface integrity.
Observeddiscrepanciesbetweenspecimenswereattributedtominorcompactionor moisture variation during curing. its strength after damage.

CHAPTER 06
CONCLUSION

Concrete remains one of the most widely used construction materials worldwide due to its affordability, strength, and adaptability. However, the growing environmental concerns associated with cement production—particularly the emission of carbon dioxide and the depletion of natural resources—have encouraged researchers to search for suitable, sustainable, and eco-friendly alternatives. This study explored the potential of agricultural waste ashes, specifically Rice Husk Ash (RHA) and Sugarcane Bagasse Ash (SCBA), as partialreplacements for cement in concrete. The research aimed to evaluate the mechanical, physical, and durability performance of concrete when cement is replaced in varying percentages by RHA, SCBA, and their combined forms.
Based on the detailed experimental investigation, the following major conclusions have been drawn:

6.1 OverallPerformanceofAgriculturalWasteAshes

The experimental results clearly demonstrate that agricultural waste ashes can significantly enhance the performance of concrete when used in controlled proportions. Both RHA andSCBA possess high silica content in amorphous form, which contributes effectively to the pozzolanic reaction. This reaction leads to the formation of additional calcium silicate hydrate (C–S–H) gel, which improves the strength and density of concrete.
The use of these ashes contributes to sustainable construction practices by transforming locally available agricultural waste into value-added construction materials. This not only reduces environmental pollution but also decreases the dependence on cement.

6.2 CompressiveStrengthBehaviour

Compressive strength testing revealed that replacing cement with RHA and SCBA resulted in noticeable improvements in the load-carrying capacity of concrete specimens, especially at higher curing periods.

Key findings include:

- At 7 days, strength increased marginally, indicating that pozzolanic activity begins early but accelerates with time.
- At 14 days, mixes containing 10% RHA and 10% SCBA outperformed the control mix due to the refinement of pores and enhancement of bondingAt 28 days, optimum strength was obtained for mixes containing: This confirms that agricultural ashes enhance the long-term strength of concretethrough secondaryhydration reactions. The microstructurebecomesmore compact, leading to increased compressive strength.

6.3 SplitTensileStrengthBehaviour

The split tensile strength values also showed substantial improvement in ash-based concrete. Although tensile strength is generally lower than compressive strength due to the brittle nature of concrete, its enhancement is an indicator of improved internal bonding.

Important observations:

- Tensilestrengthincreasedby20–30% comparedtonormal concreteforash-replaced mixes.
- The highest value was recorded for the combined ash mix (RHA5% + SCBA5%), indicating synergistic behaviour.
- RHAcontributesfineness andimproves thetransition zonebetween paste and aggregates.
- SCBAcontributesaluminaand silica,enhancingpozzolanicproperties.
- Thus,agriculturalwasteashactsas astrengtheningagent fortensileresistance.

6.4 WaterAbsorptionandPermeability

Concretedurabilityiscloselylinkedtoitswaterabsorptionrate.Lowerabsorptionindicates fewer pores and a denser structure.

Fromthe experiment:

- Allash-containingmixesabsorbedless waterthan the control mix.
- Thelowest waterabsorption valuewasobserved in thecombined ash mix.
- RHAand SCBAreducedpermeabilitydueto:
- Finerparticlesize
- Betterpackingdensity
- FormationofadditionalC–S–Hgel
- Thisimprovementimpliesbetterresistanceagainstchemicals,sulphates,andchloride attacks in aggressive environments.

6.5 WorkabilityConsiderations

- Althoughstrengthanddurabilityincreased,workabilitydecreasedslightlywiththe increase in ash content due to:
- Highersurfacearea
- Higherwaterabsorptionofashparticles
- Irregularparticleshapes
- Thiscanbecompensatedby:
- Slightadjustmentinwater–cementratio
- Useof superplasticizers
- Propermixingand curingmethods
- Thus,workabilityissuescan be easilymanagedwithout compromising performance.

6.6 MicrostructuralImprovements

- Visualfailurepatternsandqualitative examinationindicated:
- Ash-basedconcreteshowedfinercracksanddenserfracturesurfaces.
- Controlmixexhibitedwidercracks,indicatingbrittleness.
- Pozzolanicreactionimproved thebondbetween cementpasteandaggregates.
- Combinedash mixes showedthe smoothest andmost uniform texture.
- Thisprovesmicrostructuralrefinementandinternaldensificationduetotheinclusion of RHA and SCBA.

6.7 SustainabilityandEnvironmentalImpact

Theutilizationofagriculturalwasteashesplaysasignificantroleinpromotingsustainable construction:

- Cementproductionis responsiblefor around7–8%ofglobalCO_2 emissions.
- Replacingeven 10%–20%cement reduces carbon emissions considerably.
- Theproject supports waste management byusing:
- RiceHusk from ricemills
- Bagasseashfromsugarindustries
- Reduceslandfilldepositionandconvertswasteintoproductiveuse.
- Encouragesenvironmentallyresponsible engineering practices.
- Thus,theresearchstronglysupportstheadoptionofagriculturalwasteashasan environmentally sustainable material.

6.8 EconomicAdvantage

- Thestudyalsoindicatescost-savingpotential
- RHAand SCBA arelocallyavailableatminimal orzero cost.
- Reducingcementcontent lowerstheoverall costofconcrete production.
- Transportationcostsare minimizedforregions richinriceorsugarcaneindustries.
- Industrialwasteutilizationreduceswaste-handling expenses.
- Hence,theuseof agriculturalashimprovesbothaffordabilityand sustainability.

6.9 LimitationsoftheStudy

- Althoughresultswerepositive,certainlimitationswerenoted:
- Workabilitydecreaseswhen ash percentageincreases beyond the optimum level.
- Excessiveash content(>20%)mayweaken theconcrete.
- Burningtemperatureandfinenessofash significantlyaffectresults.
- Standardizationofashpreparationisstilllacking.
- Theselimitationssuggestcarefulhandlingandprocessingofashesforconsistent performance.

6.10 FinalConclusionoftheProject

Basedontheextensiveexperimentalinvestigation, itisconcluded that:

- Agriculturalwaste ashes (RHAandSCBA)areeffectivepozzolanic materials.
- 10%replacementlevelofRHAandSCBAseparatelyprovidesexcellentimprovement in strength and durability.
- Acombinedreplacementof10%(5%RHA+5%SCBA)deliversthebestoverall performance.
- Mechanicalpropertiessuchascompressiveandtensilestrengthimproved significantly.
- Durabilityparameters such as water absorption and permeabilitydecreased.
- Microstructuraldensificationwasobservedinallash-basedmixes.

FUTURE ENHANCEMENTS

7.1 Large-ScaleFieldApplications

Thecurrentstudywasconductedinacontrolledlaboratoryenvironment.However,therealtestofa sustainable material lies in its performance in actual field conditions.

Futureenhancementmayinclude:

- Conductingon-sitetrialsinvariousclimaticconditionssuchascoastalregions,humidzones,and high-temperature areas.
- UsingRHA-andSCBA-basedconcreteinpavements,low-risebuildings,boundarywalls,culverts, and rural infrastructure to evaluate its long-term performance.
- Monitoringfieldstructuresforcrackresistance,settlement,durability,andweatheringeffectovera span of 1–3 years.
- Integratingash-basedconcreteintogovernmentruraldevelopmentprojects,enablinglarge-scale sustainable construction.

7.2 DurabilityandLong-TermBehaviourStudies

Whilethe present studycovered basic durabilitytests,futureinvestigations caninclude:

- RapidChloridePenetrationTest(RCPT)toevaluatechlorideresistance.
- Sulfateattackandacidattacktests forindustrialandcoastal applications.
- Carbonationdepth analysisto studytheresistance againstatmosphericCO_2.
- Creepand shrinkagetests,especiallyforbeams andcolumns.
- Freeze–thawcycletesting, particularlyin cold-climateregions.
- Alkali–silicareaction(ASR)behaviourwhenreactiveaggregatesareused.
- Thesetestswillprovideadeeperunderstandingofthelong-termdurabilityofash-replaced concrete.

7.3 OptimizationofAshPercentageandBlendingRatios

Thestudyidentified5–10%replacementasthe optimumrange.However,further researchcan:

- ExplorecombinationslikeRHA+FlyAsh, RHA +GGBS,SCBA+Metakaolin, orTripleblends.
- Performadvancedparticle-sizeoptimizationtofurtherreducevoidsandincreasedensity.
- Analyzetheeffectofnano-silicamixedwithRHAorSCBAtoformnano-pozzolanichybrid concrete.
- Determinethemosteconomicalreplacementpercentageformass construction.

7.4 DevelopmentofAshProcessingTechnology

- TheperformanceofRHAandSCBAheavilydependsontheburning temperature,fineness, and processing method. Future enhancements should include:
- Designinglow-energyash incinerators for controlled burning at600–700°C.
- Utilizinggrindingmills toreduce ash sizetomicro/nanolevels.
- Developingportableash-processingunits forruralareas.
- Creatingindustrial standardsforashproduction, storage,and qualitycontrol.
- Thiswill help in achievingconsistent, high-qualityash suitablefor construction.

7.5 UseofAgriculturalAshinAdvancedConcreteTypes

Futurestudiescanexplorethe useofRHA andSCBA inemergingconcrete technologiessuch as:

- High-PerformanceConcrete(HPC)
- Self-CompactingConcrete (SCC)
- GeopolymerConcrete
- PerviousConcrete
- Fiber-ReinforcedConcrete
- LightweightConcrete
- Theintegrationofagriculturalashesintheseadvancedmaterialscouldopendoorstoinnovative, ultra-sustainable construction solutions.

7.6 EnvironmentalImpactandCarbonFootprintAnalysis

Althoughthe studyaddresses environmental benefits, futureresearch should:

- ConductaLife-CycleAssessment(LCA)to evaluate:
- ReductioninCO$_2$ emissions
- Reductionin energyconsumption
- Overallecologicalfootprint
- Performcarbon-neutralitycalculationsforvariousashpercentages.
- Compareash-basedconcretewithcommercial greenbinders.
- Evaluatetheimpactonwaterbodies, soilquality,andwastemanagement cycles.
- Suchassessmentswillvalidatetheenvironmentalbenefitsatnationaland global levels.

7.7 EconomicViabilityandMarketAssessment

Futureenhancementscanincludeamoredetailed economicmodel:

- Estimatingcost-benefitratiosfordifferentpercentagesofashreplacement.
- Comparingproductioncost withstandardOPC concrete.
- Assessingfinancialfeasibilityforsmall-scale ruralindustries.
- Developingbusinessmodelstoproduceandsupplyprocessed agriculturalashes.
- Studyingthepotentialformarketacceptance amongcontractorsand engineers.
- Thiswillhelpestablishagriculturalashconcreteasapracticalmaterialintheconstruction market.

7.8 StructuralBehaviourStudies

Althoughthe studymainlyfocuses on compressive and tensile strengths,futurework mayinvolve:

- Testingbeams, slabs,columns,footings, etc.
- Studyingflexuralstrength,shearstrength, andimpact resistance.
- Performingfiniteelementanalysis(FEA)tomodelstress distribution.
- Conductingnonlinear simulationto predictfailuremodes.
- Thiswill help determinethe suitabilityof ash-based concrete for load-bearing elements.

7.9 IntegrationwithSmartandGreenBuildingTechnologies

Futureresearch canalso explore:

- Usingash-basedconcrete withsolarreflectivecoatingsforlow-heat buildings.
- Combiningwith phase-changematerials (PCM) toenhancethermal comfort.
- Developingash-based 3D printingconcreteforlow-cost housing.
- Adoptingash-basedmaterialsinLEED-certified green buildings.
- Theseadvancementswillrevolutionizesustainable infrastructure.

7.10 PolicyImplementationandStandardization

Forlarge-scaleuse,government involvementisessential. Futureworkmayinclude:

- CreatingIndianStandard(IS)codeprovisionsforRHAandSCBA.
- Draftingguidelinesfor ashprocessingandmixdesign.
- Proposingpoliciesfor waste-to-resourceconversion.
- Collaboratingwithenvironmentalandagriculturaldepartments.

7.11 ScopeforAutomationandDigitalConstruction

Withthe growingtrend of automation, futureenhancements mayinclude:

- UsingAI/MLmodelsto predictash-basedconcretestrength.Developingmobileappsformix design calculation.
- Integratingash-basedconcreteintoautomatedbatching plants.
- Implementingdigital monitoringfor curingand qualitycontrol.
- Suchtechnologieswillmakesustainable constructionsmarterandmore efficient.

7.12 OverallFuturePotential

In conclusion, agricultural waste ash has enormous scope for future development. From structural applicationstoenvironmentalsustainabilityandcostreduction,thepotentialenhancementsar evast and multidimensional. With continued research, advanced technology.

LIMITATIONS

1. MaterialPreparationLimitations

1. VariationinBurningTemperature:

Theagriculturalwaste materials(RHA &SCBA) wereobtainedfrom locallyavailable sources. Theirburningtemperatureanddurationwerenotfullystandardized,whichmayhavecaused differences in ash fineness and chemical composition.

2. UncontrolledCarbon Content:

Some ash samples contained unburnt carbon,which may haveaffected pozzolanic activity and waterdemand.

3. InconsistentParticleSizeDistribution:

Despitesieving,theashparticleswerenotcompletelyuniform,whichmayinfluencethehydration process and strength development.

2. LaboratoryTestingLimitations

1. LimitedNumberofTests:

Duetotimeandresourceconstraints,onlyselectedmechanicaltests(compressivestrength,water absorption, and workability) were performed.

Otherimportanttestssuchassplittensilestrength,durabilitytests(chloridepenetration,sulphate attack, carbonation), and microstructural analysis (SEM/XRD) could not be included.

2. RestrictedCuring Conditions:

Allspecimenswerecuredundercontrolledlaboratoryconditions,whichdonotfullyrepresentreal field-based environmental variations.

3. LimitedMixProportions:

Onlyspecificreplacementpercentages(5%,10%,15%)ofRHAandSCBAwereconsidered. Larger or smaller ranges could provide deeper insights.

3. EquipmentandResource Constraints

1. Basic-LevelEquipment:

Finenesstesting,chemicalcompositionanalysis,andadvanceddurabilitytestsrequiredsophisticated equipmentthatwasnot availableinthe laboratory.

2. TimeConstraints:

Thedurationoftheprojectdidnotallowforlong-termdurabilitytestssuchas90-dayand180-day strength, drying shrinkage, or creep analysis.

3. FinancialConstraints:

Budgetlimitationsrestrictedthenumberofsamplesandnumberofrepetitionsforstatistical accuracy.

4. ScopeandDataLimitations

1. OnlyConcreteCubesTested:

Otherstructuralelementssuchasbeams,cylinders,andslabswerenotincludedinthestudy.

2. LocalMaterial Dependency:

Theresults aredependent onlocallysourced RHAand SCBA.

Ashproducedinotherregionsmaybehavedifferentlyduetovariationsinsoil,planttype,and processing method.

3. SmallSample Size:

Thelimitednumber ofconcretecubesmayintroducevariationsand reduce overall representativeness.

5. PracticalApplicationLimitations

1. Field-LevelPerformanceNotStudied:

Theprojectdoesnotincludefieldtrials,soitisdifficulttodeterminehowtheconcretewillbehave under actual environmental conditions.

2. LackofLong-TermDurabilityData:

Without extended curing and exposure testing, the long-term performance of RHA and SCBA concrete cannot be fully predicted.

3. CompatibilitywithAdmixturesUnknown:

Thestudydoesnotanalyzetheinteractionofagriculturalwasteashwithcommercialchemical admixtures like superplasticizers, retarders, or accelerators.

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